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ORIGINAL RESEARCH

Multi-agent dynamic financial portfolio management: a differential game approach

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Abstract
In this paper we extend the classical Merton model to the case of several players and we propose a multi-agent Merton-type portfolio optimization model formulated by means of differential games. Each player is price-taker and invests on the market in order to maximize her inter-temporal utility. On the market there are p different assets, $p-1$ of them are risky and their price are generated by a geometric Brownian motion, while one is risk-free. We discuss the non-cooperative and cooperative case and show the existence of a closed-form solution when the class of CRRA utility functions is assumed for both infinite and finite time horizon. We perform a numerical study which illustrates that a cooperative strategy is preferable to a non-cooperative one as it allows for the adoption of optimal strategies from both players. Non-cooperation, instead, leads to sub-optimal solutions.

Keywords Differential games · Dynamic games · Dynamic programming · Merton's portfolio model · Noncooperative versus cooperative games

JEL Classification G11 · C70

1 Introduction

During the last century the theory of portfolio management has been enriched with several contributions and developments. The classical portfolio model by Markowitz (1952), in which a Decision Maker (DM) selects the most efficient portfolio through different possible combinations of different investment strategies, is the most important in this vein (Aouni et al., 2014, 2012; Benson, 1985; Ben Abdelaziz et al., 1999; Ben Abdelaziz and Masri, 2005; Ben Abdelaziz et al., 2020; La Torre and Mendivil, 2018).

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The Markowitz model has notable drawbacks, including its limited ability to capture the uncertainty associated with portfolio decisions and its static formulation that does not consider the possibility of portfolio rebalancing over the investment horizon.

On one hand, the literature proposes various extended portfolio models to address multiple and conflicting criteria, as well as the concepts of uncertainty and incomplete information. In practical applications, verifying the assumptions required to determine a single optimal solution is often challenging. Specifically, most of the time, it is difficult to explicitly define the utility function because numerous unpredictable factors can influence decision-makers. These situations can be represented by incorporating uncertainty or incomplete information into the utility function (Ben Abdelaziz and Masri, 2005; La Torre and Mendivil, 2018).

On the other hand, several contributions consider dynamic portfolio allocation models, and among them, the most famous is the one introduced in 1969 by Merton (1969) [see also (Merton, 1971, 1987)] in his famous article, *"Lifetime Portfolio Selection Under Uncertainty: The Continuous-Time Case"*. The portfolio model introduced in this article, known later on as Merton's model, is a continuous-time problem in which the investor has to decide the allocation of her wealth between a risky asset (later, the problem has been extended to the case with multiple risky assets) and a risk-free one with the goal of maximising her intertemporal utility. In the classic model proposed by Merton, some assumptions are introduced to simplify the model and make it more tractable from a mathematical perspective. The first hypothesis considers that all the assets among which it is possible to invest are of the limited liability (the equity holders have the right, not the obligation, to pay off the debt holders and take over the remaining assets of the firm) type and the stock prices are governed by Geometric Brownian Motion processes. The second hypothesis assumes a continuously-trading market with no transaction costs. Finally, the last assumption considers that there is no default risk for the issuer of the financial assets. In Merton (1969), the optimal allocation problem is presented in two different possible scenarios: the *finite-horizon* case with $t \in [0, T]$, and the *infinite-horizon* case with $t \in [0, +\infty)$.

There are several advantages in utilizing a dynamic Merton's portfolio allocation versus a static Markowitz approach. A dynamic framework refers to the practice of actively adjusting the allocation of assets within a portfolio over time based on changing market conditions, while static portfolio allocation involves maintaining a fixed allocation of assets without making frequent adjustments. Therefore, a dynamic approach shows adaptability to changing market conditions and allows investors to respond to evolving market conditions, economic factors, and financial trends. It also enables them to adjust their portfolios in real time to capitalize on opportunities and mitigate risks. In contrast, static portfolio allocation maintains a fixed allocation, which may not be optimal in varying market environments. Another interesting advantage is related to risk management: dynamic portfolio allocation allows for better risk management by actively responding to market volatility. It enables investors to reduce exposure to high-risk assets during periods of heightened market turbulence and increase exposure to less risky assets. This flexibility helps minimize potential losses and preserve capital. A dynamic formulation also has the potential for enhanced returns: by actively adjusting the portfolio based on market conditions, dynamic portfolio allocation seeks to identify and seize investment opportunities that can potentially generate higher returns. It may involve reallocating investments to sectors or asset classes that are expected to outperform in the current market environment. In contrast, static portfolio allocation may limit the ability to capture such opportunities. Dynamic portfolio allocation can optimize diversification by identifying and adjusting the portfolio's asset allocation based on the correlations between different asset classes. It allows for rebalancing the portfolio to maintain an optimal diversification level, which can help reduce the overall risk and enhance potential returns.

Static portfolio allocation may not adapt to changes in asset correlations, potentially resulting in suboptimal diversification. Finally, a dynamic portfolio allocation requires active management and monitoring of market conditions. This active approach allows investors to take advantage of market inefficiencies, emerging trends, and new information. It enables them to make timely investment decisions and adjust the portfolio to align with their investment objectives. On the other hand, static portfolio allocation relies on a passive approach without considering real-time market dynamics.

There are two potential approaches to extending Merton's model: one involves utilizing dynamic multiobjective optimization techniques, while the other involves applying a dynamic game framework.

Dynamic multiobjective models in portfolio management use optimization techniques to generate a dynamic set of Pareto-optimal portfolios. These portfolios represent different combinations of asset allocations that achieve the best trade-offs between risk and return, catering to the preferences of different investors. Therefore, investors can select, at each instant of time, a portfolio from the dynamic Pareto front based on their risk appetite and return requirements, effectively making a decision based on their preferences.

In the dynamic game approach, instead, the model focuses on capturing the decision-making processes and interactions of individual agents within the market. Each agent makes investment decisions independently based on their own objectives and market information. The collective behaviour of these agents influences the overall market dynamics and portfolio performance. They aim to understand how the behaviour and strategies of individual agents impact the overall market dynamics in both cooperative and non-cooperative frameworks, including asset prices, trading volumes, and portfolio performance.

In this paper, we propose an extension of Merton's model under a dynamic game approach to the case of several agents but outside the context of the price equilibrium. More in detail, we consider the case of two players acting on the same financial market and sharing the possibility of investing their wealth in a fixed and common number of possible risky and risk-free assets. We first focus on the infinite-horizon formulation, and then we discuss how to modify our conclusions in the finite-horizon case. In both cases, we determine the optimal level of consumption and the optimal portfolio allocation across different assets. In fact, under the hypothesis of considering a Constant Relative Risk Aversion (CRRA) utility function, it is possible to determine the closed-form solution of the Bellman equation and, therefore, the optimal time-continuous paths.

The paper is organized as follows. Sect. 2 presents a literature review on the several extensions of Merton's model while Sect. 3 summarizes the classic Merton's model for an agent in continuous time. In Sect. 4, we formulate our multi-agent Merton-type model under a dynamic game approach. Section 5 is dedicated primarily to the analysis of the non-cooperative case and then of the cooperative one, considering the CRRA utility function as a power function. The logarithm expression of the CRRA utility is used in Sect. 6, and similar investigations to the previous section are conducted. Section 7 shows how the previous results would become in the case of the finite-horizon model. In Sect. 8, we present a case study where we compare the implementation of non-cooperative versus cooperative strategies for a simple dynamic portfolio model. Sect. 9 discusses the managerial implications of our model, and Sect. 10 concludes the paper.

2 Literature review

A number of extensions of Merton's model have incorporated more complex features into the models, such as bankruptcy [see, e.g., (Karatzas et al., 1986; Sethi and Taksar, 1992)] or general price processes [e.g., (Karatzas et al., 1987; Karatzas and Žitković, 2003)]. In Karatzas et al. (1986), the authors determined the single-agent optimal policies and the value function by using weaker assumptions on the utility function than the ones required on Merton (1969). Moreover, they removed the restriction $U'(0) = \infty$ used to guarantee that, under an optimal policy, the consumption constraint will not be active and bankruptcy will not occur. Hence, they carefully treated the consumption constraint and addressed the possibility of bankruptcy. In Karatzas et al. (1987) instead, it was considered a martingale-based approach to the problem in complete Itô process markets. This approach applies to a much more general class of stock price processes, including non-Markovian models, and was used to explicitly characterise single-agent optimal policies for general utility functions and semi-martingale stock price processes. Other works included multiple agents in the model, such as Karatzas et al. (1990) and Tapiero (2015), but they did so in an equilibrium context; each agent's behaviour depends on the others only through the price, which is determined in equilibrium. In Karatzas et al. (1990), the multi-agent consumption/investment problem was reduced to the problem of finding an appropriate "representative agent" who represents the individual interests of the many agents and has their aggregate income. In this setting, the authors proved the existence and uniqueness of equilibrium prices and characterized these prices as well as the consumption and investment decisions made by the individual agents. Tapiero (2015) considers a wealth heterogeneous multi-agent financial Consumption Capital Assets Pricing Model (CCAPM), where instead of defining a representative agent underlying the pricing model, assumes that each agent is, in fact, 'Cournot-gaming' a market defined by all other agents. This results in a decomposition of an n-agents game into n games of two agents: a specific agent and a synthetic one (a proxy for all other agents), on the basis of which an equilibrium consumption price solution is defined. More recently, Kogan and Tapiero (2018) proposed the formulation of the multi-agent financial Merton's model as a stochastic differential Cournot game, resulting in a unique consumption price as a function of aggregate demand for consumption and supply factors. More specifically, the authors defined the "Cournot game" as that of an individual agent "gaming the market" and pointed out that consumption decisions depend on consumers' strategic strengths. A mean field approach to the optimal consumption/investment problem under relative performance criteria was proposed by Lacker and Soret (2020), where a closed-form solution for the n-player game and the corresponding mean field game is provided. That solution is unique in the class of equilibria with constant investment and continuous time-dependent consumption, both independent of the wealth of the agent. Other extensions can be found, for instance, in Ben Abdelaziz and La Torre (2021), Chang (2004) and Hamacher et al. (2007).

3 Preliminaries

Merton's model is an inter-temporal portfolio choice problem. In its classical formulation, it is defined as a stochastic optimal control model, and it can be solved in a closed-form solution by means of the Hamilton–Jacobi–Bellman (HJB) equation. Analytical tractability in Merton's classical formulation is due to the following conditions: normal distribution of

asset returns, constant relative risk-aversion, continuous trading. Moreover, Merton's model formulation is based on the following assumptions:

- current wealth is $W_0 > 0$, and there is an infinite life expectation;
- it is possible to invest in (allocate to) $p-1$ risky assets and a risk-free asset;
- each risky asset has known normal distribution of returns;
- it is allowed to long or short any fractional quantities of assets;
- trading is done in continuous time $t \geq 0$, with no transaction costs;
- one can consume any fractional amount of wealth at any time;
- decisions are taken in a dynamic framework, and both the share of wealth invested in the $i-$th asset s_i, for every asset i, and the consumption c can be done at each time;
- the decision maker aims at maximizing the lifetime-aggregated utility of consumption;
- consumption utility $U : \mathbb{R}^+ \to \mathbb{R}$ is a concave C^2 (twice differentiable with continuous derivatives) function and is assumed to have constant relative risk-aversion.

Therefore, Merton's model reads as:

$$\max_{c,s_1,s_2,\ldots,s_{p-1}} J(W, c, s_1, s_2, \ldots, s_{p-1}) := \mathbb{E}\int_0^{+\infty} e^{-\rho t} U(c(t))dt \tag{1}$$

Subject to:

$$\begin{cases} dW(t) = \left[\left(\sum_{i=1}^{p-1} s_i(t)(\mu_i - \mu_p) + \mu_p\right) W(t) - c(t)\right] dt + W(t)\sum_{i=1}^{p-1} s_i(t)\sigma_i dB_i(t) \\ W(0) = W_0 > 0 \text{ given}, \\ W(t) > 0 \ \forall t > 0, \\ dP_i(t) = P_i(t)(\mu_i dt + \sigma_i dB_i(t)), \qquad P_i(0) > 0 \text{ given}, \quad i = 1, \ldots, p-1 \\ dP_p(t) = \mu_p P_p(t), \qquad P_p(0) > 0 \text{ given}, \\ \sum_{i=1}^{p} s_i(t) = 1, \\ 0 \leq s_i(t) \leq \xi_i, \quad i = 1, \ldots, p \end{cases} \tag{2}$$

This is as a continuous-time stochastic control problem in which the state at time t is $W(t)$, the control at the time t is described by $s_i(t)$ and $c(t)$, the reward per unit of time is $U(c(t))$, and the return at the time t is the accumulated discounted reward:

$$\int_t^{+\infty} e^{-\rho(s-t)} U(c(s))ds, \tag{3}$$

where $\rho > 0$ is a discount factor. The price of the i-th risky asset, denoted by $P_i(t)$, $i = 1, \ldots, p-1$, is modelled by the following stochastic process (geometric Brownian motion):

$$dP_i(t) = P_i(t)(\mu_i dt + \sigma_i\, dB_i(t)), \quad i = 1, \ldots, p-1 \tag{4}$$

where μ_i is the instantaneous conditional expected rate of P_i per unit time, $\sigma_i > 0$ is the instantaneous conditional covariance per unit time and $dB_i(t)$ is a Wiener process independent from each others for every i. The p-th asset is purely deterministic (risk-free); therefore, its price $P_p(t)$ is generated by:

$$dP_p(t) = \mu_p(t)P_p(t)dt, \tag{5}$$

where μ_p stands for the risk-free interest rate. Since $s_i(t)$ is the share of wealth invested in the $i-th$ asset, follows that

$$\sum_{i=1}^{p} s_i(t) = 1. \tag{6}$$

The total wealth $W(t)$ is defined to be

$$W(t) = \sum_{i=1}^{p} N_i(t) P_i(t), \tag{7}$$

where $N_i(t)$ are the units of the asset i owned by the investor at time t. Note that, since $N_i(t)$ can be expressed as $N_i(t) = s_i(t)W(t)/P_i(t)$, then from (7) follows (6). By differentiating (7) and considering the expression of $N_i(t)$ (in terms of $s_i(t)$ and $P_i(t)$) as well as (4), (5) and the consumption c, we get the following budget equation [see (Merton, 1969)]:

$$dW(t) = \left[\left(\sum_{i=1}^{p-1} s_i(t)(\mu_i - \mu_p) + \mu_p \right) W(t) - c(t) \right] dt + \sum_{i=1}^{p-1} s_i(t)\sigma_i W(t) dB_i(t).$$

Finally, we denote by $\xi_i \in (0, 1]$ the weight in terms of volume of the asset i on the total market volume.

The financial portfolio manager aims to find the optimal investment policy that maximizes the inter-temporal utility. In order to do this, she will decide the optimal share of wealth to invest in the i-th asset by taking into account the volatility of each risky asset and the overall level of consumption.

4 Main result: a multi-agent dynamic portfolio model

The aim of this section is to present our main model formulation. Our goal is to extend the classical Merton's model with a single player to the case of two players that invest in the same financial market and analyze cooperative vs non-cooperative strategies. As described in the sequel, our analysis confirms that cooperation is the best strategy as, in this scenario, both players can put in place their optimal investment plans. In this scenario, indeed, the two decision-makers share their information about financial market risk, and this allows us to assume that both of them have the same risk aversion parameter. Non-cooperation, instead, leads to the implementation of sub-optimal strategies. Our formulation takes the form of a dynamic game in which each of the two players aims to solve the following problem:

$$\max_{c^j, s_1^j, s_2^j, \ldots, s_{p-1}^j} J(W^j, c^j, s_1^j, s_2^j, \ldots, s_{p-1}^j) := \mathbb{E} \int_0^{+\infty} e^{-\rho t} \sum_{j=1}^{2} U_j(c^1(t), c^2(t)) dt \tag{8}$$

Subject to:

$$\begin{cases} dW^j(t) = \left[\left(\sum_{i=1}^{p-1} s_i^j(t)(\mu_i - \mu_p) + \mu_p\right) W^j(t) - c^j(t)\right] dt + W^j(t) \sum_{i=1}^{p-1} s_i^j(t)\sigma_i dB_i(t) \\ W^j(0) = W^j > 0 \text{ given}, \\ W^j(t) > 0 \ \forall t \in (0, T], \\ dP_i(t) = P_i(t)(\mu_i\, dt + \sigma_i\, dB_i(t)), \qquad P_i(0) > 0 \text{ given}, \quad i = 1, \ldots, p-1 \\ dP_p(t) = \mu_p P_p(t) dt, \qquad P_p(0) > 0 \text{ given}, \\ \sum_{i=1}^{p} s_i^j(t) = 1, \qquad j = 1, 2 \\ 0 \le \sum_{j=1}^{2} s_i^j(t) \le \xi_i, \qquad i = 1, \ldots, p \end{cases} \tag{9}$$

Most of the parameters and the equations listed in (8)–(9) can be seen as a natural extension of those presented in the previous section for a single decision maker. However, the following considerations hold true:

- Both the two players, $j = 1, 2$, invest on the financial market and share the same number of investments. There are p available assets, $p - 1$ are risky and one is risk-free.
- s_i^j is the share of wealth of the player j invested in the i-th asset and hence

$$\sum_{i=1}^{p} s_i^j(t) = 1. \tag{10}$$

  Moreover, we can establish the following relation:

$$0 \le \sum_{j=1}^{2} s_i^j(t) \le \xi_i, \tag{11}$$

  which states that the sum of the shares of wealth of both players invested in the same asset i will at most be equal to the availability, in terms of volume, of the asset i on the total market volume. Note that (10) and (11) imply the following technical condition:

$$\sum_{i=1}^{p} \xi_i \ge 2, \tag{12}$$

  because $0 \le s_i^j(\cdot) \le 1$ for $i = 1, \ldots, p$, and $j = 1, 2$. Then, the sum of the upper bounds has to be greater or equal to 2. Moreover, (11) will take the form of an equality, i.e.,

$$\sum_{j=1}^{2} s_i^j(t) = \xi_i, \tag{13}$$

  in the cooperative framework as it represents the external enforcement that allows the players to cooperate.
- $U_j : \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}$, $j = 1, 2$, is a concave C^2 utility function with respect to both variables. Note that each player's utility function also depends on the consumption of the other, which is reasonable as the two players operate in the same financial market.

In the next sections, we discuss how to obtain optimal strategies for players in both cooperative and non-cooperative contexts. We consider both the finite and the infinite time horizon, and we determine the optimal solution to the HJB equation for power and logarithmic utilities.

5 The case of power utility

5.1 The non-cooperative case

In a non-cooperative framework, the player 1 (the player 2) solves the problem (8)–(9) by taking the behaviour of the player 2 (player 1) as given, meaning that each player cares only about its investments on the market. Hence, for the player 1, solving the problem (8) is equivalent to solve another maximization problem in which the utility function U_1 only depends on $c^1(t)$, as $c^2(t)$ can be considered to be exogenous and, therefore, it will not affect the computation of the maximum. Moreover, the player 1 also know the shares of wealth of the player 2 invested in all assets i, as this data is provided exogenously.

Similar considerations also hold for the player 2. Then, it is possible to show that the following result for the player 1 holds true. An equivalent result also holds for the player 2 (see Remark 1).

Proposition 1 *Let us suppose that:*

$$\rho > (1-\alpha_1)\left(\mu_p + \frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i-\mu_p)^2}{\alpha_1(\sigma_i)^2}\right)$$

and assume that the utility function takes the form:

$$U_1(c^1(t), c^2(t)) = \frac{1}{1-\alpha_1}(c^1(t))^{1-\alpha_1} + \frac{1}{1-\alpha_2}(c^2(t))^{1-\alpha_2},$$

with constant relative risk aversion $\alpha_j > 0$, $\alpha_j \neq 1$ for $j = 1, 2$, and α_1 such that $0 \leq s_i^1 < \xi_i - s_i^2$. Then the non-cooperative optimal consumption and investment on the i-th asset of the player 1, along with her total wealth are respectively given by:

$$c^1(t) = \left(\frac{\rho}{\alpha_1} - \frac{1-\alpha_1}{\alpha_1}\left[\mu_p + \frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i-\mu_p)^2}{\alpha_1(\sigma_i)^2}\right]\right) W^1(t), \tag{14}$$

$$s_i^1(t) = \frac{\mu_i-\mu_p}{\alpha_1(\sigma_i)^2}, \quad i = 1, \ldots, p-1. \tag{15}$$

$$W^1(t) = W^1(0)\exp\left[\left(\frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i-\mu_p)^2}{\alpha_1(\sigma_i)^2} + \frac{\mu_p-\rho}{\alpha_1}\right)t + \sum_{i=1}^{p-1}\frac{\mu_i-\mu_p}{\alpha_1\sigma_i}(B_i(t)-B_i(0))\right]. \tag{16}$$

Proof Denoting by $V_1 \in C^2$ the value function associated to the problem (8)–(9) for the player 1, the triple $(W^1(t), c^1(t), s_i^1(t))$ is its solution if $V_1(W^1(t))$ solves the following Hamilton–Jacobi–Bellman (HJB) equation:

$$\begin{aligned} -\rho V_1 + \max_{\{c^1, s_i^1\}}\Bigg\{&\left[\left(\sum_{i=1}^{p-1} s_i^1(t)(\mu_i-\mu_p) + \mu_p\right)W^1(t) - c^1(t)\right]D_{W^1}V_1 + U_1(c^1(t), c^2(t)) \\ &+ \frac{1}{2}\sum_{i=1}^{p-1}(s_i^1(t))^2(\sigma_i)^2(W^1(t))^2 D^2_{W^1}V_1\Bigg\} = 0, \end{aligned} \tag{17}$$

where $D_{W^1}V_1$ is the first derivative of the value function V_1 with respect to W^1, while $D^2_{W^1}V_1$ is its second derivative with respect to W^1. Denoting by $H_1(W^1, c^1, s^1_i, D_{W^1}V_1, D^2_{W^1}V_1)$ the Hamiltonian associated to (17), and assuming that

$$U_1(c^1(t), c^2(t)) = \frac{1}{1-\alpha_1}(c^1(t))^{1-\alpha_1} + \frac{1}{1-\alpha_2}(c^2(t))^{1-\alpha_2},\ \alpha_j > 0,\ \alpha_j \neq 1,\quad j = 1, 2, \tag{18}$$

$$V_1(W^1(t)) = \frac{A_1^{-\alpha_1}}{1-\alpha_1}(W^1(t))^{1-\alpha_1},\ A_1 \in \mathbb{R},\ \alpha_1 > 0,\ \alpha_1 \neq 1, \tag{19}$$

we apply the first-order conditions as to get:

$$\frac{\partial H_1}{\partial c^1(t)} = -A_1^{-\alpha_1}(W^1(t))^{-\alpha_1} + (c^1(t))^{-\alpha_1} = 0 \iff c^1(t) = A_1 W^1(t), \tag{20}$$

$$\begin{aligned}\frac{\partial H_1}{\partial s^1_i(t)} &= \sum_{i=1}^{p-1}(\mu_i - \mu_p)A_1^{-\alpha_1}(W^1(t))^{1-\alpha_1} + \sum_{i=1}^{p-1}(s^1_i(t))(\sigma_i)^2 A_1^{-\alpha_1}(-\alpha_1)(W^1(t))^{1-\alpha_1} = 0\\ &\iff s^1_i(t) = \frac{\mu_i - \mu_p}{\alpha_1(\sigma_i)^2},\quad i = 1, \ldots, p-1,\end{aligned} \tag{21}$$

Note that the condition $0 \leq s^1_i < \xi_i - s^2_i$ that follows from (11) in the non-cooperative framework, is satisfied for every $i = 1, \ldots, p-1$, if

$$\alpha_1 > \max\left\{\frac{\mu_1 - \mu_p}{(\sigma_1)^2(\xi_1 - s^2_1)}, \frac{\mu_2 - \mu_p}{(\sigma_2)^2(\xi_2 - s^2_2)}, \ldots, \frac{\mu_{p-1} - \mu_p}{(\sigma_{p-1})^2(\xi_{p-1} - s^2_{p-1})}\right\}, \tag{22}$$

with $\mu_i - \mu_p \geq 0, \forall\, i = 1, \ldots, p-1$.

By plugging the expressions of (18), (19), (20), (21) into (17) and remembering that the utility U_1 only depends on the $c^1(t)$ term (due the equivalence of the maximization problems observed at the begin of the section) we get

$$A_1 = \frac{\rho}{\alpha_1} - \frac{1-\alpha_1}{\alpha_1}\left[\mu_p + \frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i - \mu_p)^2}{\alpha_1(\sigma_i)^2}\right]. \tag{23}$$

To ensure that V_1 is a concave function, it is necessary that $A_1 > 0$ and hence that

$$\rho > (1-\alpha_1)\left(\mu_p + \frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i - \mu_p)^2}{\alpha_1(\sigma_i)^2}\right).$$

Then, by replacing (23) into (20) and using (21) we get the closed-form expressions of the controls given by (14) and (15). By using (14) and (15) in the first equation of (9), the optimal wealth of the player 1 follows:

$$W^1(t) = W^1(0)\exp\left[\left(\frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i-\mu_p)^2}{\alpha_1(\sigma_i)^2}+\frac{\mu_p-\rho}{\alpha_1}\right)t+\sum_{i=1}^{p-1}\frac{\mu_i-\mu_p}{\alpha_1\sigma_i}\left(B_i(t)-B_i(0)\right)\right]. \tag{24}$$

The transversality condition reads as follows:

$$\lim_{t\to+\infty} e^{-\rho t}\,\mathbb{E}\left[V_1(W^1(t))\right] = \lim_{t\to+\infty} e^{-\rho t}\cdot\frac{A_1^{-\alpha_1}}{1-\alpha_1}\mathbb{E}\left[(W^1(t))^{1-\alpha_1}\right] = 0, \tag{25}$$

which is verified noting that by Jensen's inequality $\mathbb{E}\left[(W^1(t))^{1-\alpha_1}\right] \le \mathbb{E}\left[(W^1(t))\right]^{1-\alpha_1}$ and

$$\begin{aligned}&\lim_{t\to+\infty} e^{-\rho t}\cdot\frac{A_1^{-\alpha_1}}{1-\alpha_1}\mathbb{E}\left[(W^1(t))\right]^{1-\alpha_1}\\ &= \lim_{t\to+\infty}\frac{A_1^{-\alpha_1}}{1-\alpha_1}\left(W^1(0)\exp\left[\left(-\frac{\rho}{1-\alpha_1}+\frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i-\mu_p)^2}{\alpha_1(\sigma_i)^2}+\frac{\mu_p-\rho}{\alpha_1}\right)t\right]\right)^{1-\alpha_1} = 0\end{aligned} \tag{26}$$

if and only if

$$\rho > (1-\alpha_1)\left(\mu_p+\frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i-\mu_p)^2}{(\sigma_i)^2}\right). \tag{27}$$

Therefore, since

$$0 \le e^{-\rho t}\cdot\frac{A_1^{-\alpha_1}}{1-\alpha_1}\mathbb{E}\left[(W^1(t))^{1-\alpha_1}\right] \le e^{-\rho t}\cdot\frac{A_1^{-\alpha_1}}{1-\alpha_1}\mathbb{E}\left[(W^1(t))\right]^{1-\alpha_1},$$

by the limit comparison theorem for $t \to +\infty$ follows (25). □

Remark 1 As anticipated at the beginning of the section, a result equivalent to Proposition 1 holds for player 2. Indeed, proceeding as in the proof of Proposition 1, we get that the non-cooperative optimal consumption and the investment on the i-th asset together with the total wealth of the player 2 are given by:

$$\begin{aligned}c^2(t) &= \left(\frac{\rho}{\alpha_2}-\frac{1-\alpha_2}{\alpha_2}\left[\mu_p+\frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i-\mu_p)^2}{\alpha_2(\sigma_i)^2}\right]\right)W^2(t),\\ s_i^2(t) &= \frac{\mu_i-\mu_p}{\alpha_2(\sigma_i)^2},\quad i=1,\dots,p-1,\\ W^2(t) &= W^2(0)\exp\left[\left(\frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i-\mu_p)^2}{\alpha_2(\sigma_i)^2}+\frac{\mu_p-\rho}{\alpha_2}\right)t+\sum_{i=1}^{p-1}\frac{\mu_i-\mu_p}{\alpha_2\,\sigma_i}\left(B_i(t)-B_i(0)\right)\right].\end{aligned}$$

Remark 2 Note that, as will be shown in the numerical study in Sect. 8, in the non-cooperative case, the use of both optimal strategies by the two players may not be admissible due to the lack of the necessary volume of assets on the market.

5.2 The cooperative case

Now, we focus on the cooperative setup where the two players agree on the investments to be made on the market so that both maximize their intertemporal utility. Moreover, they share the same constant relative risk aversion, i.e., $\alpha_1 = \alpha_2 = \alpha$.

Then, the following holds.

Proposition 2 *Suppose that*

$$\rho > (1-\alpha)\left[\mu_p + \frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i - \mu_p)\right],$$

and the utility functions U_1 and U_2 in (8) take the same form:

$$U_1(c^1(t), c^2(t)) = U_2(c^1(t), c^2(t)) = \frac{1}{1-\alpha}\left[(c^1(t))^{1-\alpha} + (c^2(t))^{1-\alpha}\right],$$

where $\alpha > 0$, $\alpha \neq 1$ is the common constant relative risk aversion. Then the cooperative optimal controls for the two players $j = 1, 2$, along with their total wealth are respectively given by:

$$c^j(t) = \left(\frac{\rho}{\alpha} - \frac{1-\alpha}{\alpha}\left[\mu_p + \frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i - \mu_p)\right]\right) W^j(t), \tag{28}$$

$$s_i^j(t) = \frac{\xi_i}{2}, \quad i = 1, \ldots, p-1, \tag{29}$$

$$W^j(t) = W^j(0)\exp\left[\left(\frac{\mu_p - \rho}{\alpha} + \frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i - \mu_p)\right)t + \frac{1}{2}\sum_{i=1}^{p-1}\sigma_i\,\xi_i\,(B_i(t) - B_i(0))\right] \tag{30}$$

Proof By defining $V \in C^2$ the value function associated with the problem (8)–(9), the triple $(W^j(t), c^j(t), s_i^j(t))$ for $j = 1, 2$ is its solution if $V(W^1(t), W^2(t))$ solves the following HJB equation:

$$\begin{aligned} -\rho V + \max_{\{c^1, c^2, s_i^1, s_i^2\}} &\left\{\sum_{j=1}^{2}\left\langle\left(\sum_{i=1}^{p-1} s_i^j(t)(\mu_i - \mu_p) + \mu_p\right)W^j(t) - c^j(t), D_{W^j}V\right\rangle + U_1(c^1(t), c^2(t))\right. \\ &\left. + U_2(c^1(t), c^2(t)) + \frac{1}{2}\sum_{j,k=1}^{2}\left(\sum_{i=1}^{p-1}(s_i^j(t))(s_i^k(t))(\sigma_i)^2(W^j(t))(W^k(t))D^2_{W^jW^k}V\right)\right\} = 0, \end{aligned} \tag{31}$$

where $D_{W^j}V$ is the first derivative of the value function V with respect to W^j, while $D^2_{W^jW^k}V$ is its second derivative with respect to W^jW^k for $j, k = 1, 2$.

Let $H(W, c, s_i, D_W V, D^2_W V)$ be the Hamiltonian associated to (31), where

$$W = [W^1, W^2], \quad c = [c^1, c^2], \quad s_i = [s_i^1, s_i^2], \quad D_W V = [D_{W^1}V, D_{W^2}V],$$

and

$$D^2_W V = \begin{pmatrix} D^2_{(W^1)^2}V & D^2_{W^1W^2}V \\ D^2_{W^2W^1}V & D^2_{(W^2)^2}V \end{pmatrix}.$$

Assuming that

$$U_1(c^1(t), c^2(t)) = U_2(c^1(t), c^2(t)) = \frac{1}{1-\alpha}\left[(c^1(t))^{1-\alpha} + (c^2(t))^{1-\alpha}\right],\ \alpha > 0,\ \alpha \neq 1, \tag{32}$$

$$V(W^1(t), W^2(t)) = \frac{A^{-\alpha}}{1-\alpha}\left[(W^1(t))^{1-\alpha} + (W^2(t))^{1-\alpha}\right],\ A \in \mathbb{R},\ \alpha > 0,\ \alpha \neq 1, \tag{33}$$

we can apply the first order conditions to get the expression of the controls:

$$\frac{\partial H}{\partial c^j(t)} = -A^{-\alpha}(W^j(t))^{-\alpha} + 2(c^j(t))^{-\alpha} = 0 \iff c^j(t) = 2^{1/\alpha} A W^j(t), \quad j = 1, 2 \tag{34}$$

$$\frac{\partial H}{\partial s_i^j(t)} = \sum_{i=1}^{p-1}(\mu_i - \mu_p)A^{-\alpha}(W^j(t))^{1-\alpha} + \sum_{i=1}^{p-1}(s_i^j(t))(\sigma_i)^2 A^{-\alpha}(-\alpha)(W^j(t))^{1-\alpha} = 0$$
$$\iff s_i^j(t) = \frac{\mu_i - \mu_p}{\alpha(\sigma_i)^2}, \quad j = 1, 2, \quad i = 1, \ldots, p-1. \tag{35}$$

By plugging the expressions of (32), (33), (34) and (35) into (31) follows that

$$A = \frac{\rho}{\alpha\, 2^{1/\alpha}} - \frac{1-\alpha}{\alpha\, 2^{1/\alpha}}\left[\mu_p + \frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i - \mu_p)^2}{\alpha(\sigma_i)^2}\right]. \tag{36}$$

Finally, imposing to (36) the constraint (13) that, using (35) can be re-written as

$$\frac{\mu_i - \mu_p}{\alpha(\sigma_i)^2} = \frac{\xi_i}{2}, \qquad i = 1, \ldots, p-1 \tag{37}$$

we obtain

$$A = \frac{\rho}{\alpha\, 2^{1/\alpha}} - \frac{1-\alpha}{\alpha\, 2^{1/\alpha}}\left[\mu_p + \frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i - \mu_p)\right]. \tag{38}$$

We require that A (38) be strictly positive to guarantee the concavity of V, and hence that

$$\rho > (1-\alpha)\left[\mu_p + \frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i - \mu_p)\right]. \tag{39}$$

Then, by replacing (38) into (34) and rewriting (35) using (37), we get (28) and (29). The optimal wealth of the player j, $j = 1, 2$, i given by:

$$W^j(t) = W^j(0)\exp\left[\left(\frac{\mu_p - \rho}{\alpha} + \frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i - \mu_p)\right)t + \frac{1}{2}\sum_{i=1}^{p-1}\sigma_i\,\xi_i\,(B_i(t) - B_i(0))\right]. \tag{40}$$

The transversality condition is given by:

$$\lim_{t\to+\infty} e^{-\rho t}\,\mathbb{E}\left[V(W^1(t), W^2(t))\right] = \lim_{t\to+\infty} e^{-\rho t}\cdot\frac{A^{-\alpha}}{1-\alpha}\left(\mathbb{E}\left[(W^1(t))^{1-\alpha} + (W^2(t))^{1-\alpha}\right]\right) = 0, \tag{41}$$

which is satisfied, noting that, by Jensen's inequality follows

$$\mathbb{E}\left[(W^1(t))^{1-\alpha}+(W^2(t))^{1-\alpha}\right]\leq \mathbb{E}\left[(W^1(t))\right]^{1-\alpha}+\mathbb{E}\left[(W^2(t))\right]^{1-\alpha},$$

and

$$\lim_{t\to+\infty} e^{-\rho t}\cdot\frac{A^{-\alpha}}{1-\alpha}\left(\mathbb{E}\left[(W^1(t))\right]^{1-\alpha}+\mathbb{E}\left[(W^2(t))\right]^{1-\alpha}\right)=$$
$$\lim_{t\to+\infty}\frac{A^{-\alpha}}{1-\alpha}\left[\left(W^1(0)\exp\left[\left(-\frac{\rho}{1-\alpha}+\frac{\mu_p-\rho}{\alpha}+\frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i-\mu_p)\right)t\right]\right)^{1-\alpha}+\right.$$
$$\left.\left(W^2(0)\exp\left[\left(-\frac{\rho}{1-\alpha}+\frac{\mu_p-\rho}{\alpha}+\frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i-\mu_p)\right)t\right]\right)^{1-\alpha}\right]=0$$

if and only if

$$\rho>(1-\alpha)\left[\mu_p+\frac{\alpha}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i-\mu_p)\right] \tag{42}$$

Therefore, since

$$0\leq e^{-\rho t}\cdot\frac{A^{-\alpha}}{1-\alpha}\mathbb{E}\left[(W^1(t))^{1-\alpha}+(W^2(t))^{1-\alpha}\right]\leq e^{-\rho t}\cdot\frac{A^{-\alpha}}{1-\alpha}\left(\mathbb{E}\left[(W^1(t))\right]^{1-\alpha}+\mathbb{E}\left[(W^2(t))\right]^{1-\alpha}\right),$$

by the limit comparison theorem for $t\to+\infty$ follows (41). □

6 The case of log utility

In this section, following the same reasoning as in Sect. 5, we first consider the non-cooperative case and then the cooperative one. The following result holds for the player 1, which solves (8)–(9) by knowing the behaviour of the player 2.

Proposition 3 *If the utility function takes the form*

$$U_1(c^1(t),c^2(t))=\ln(c^1(t))+\ln(c^2(t)), \tag{43}$$

and s_i^1 is such that $0\leq s_i^1<\xi_i-s_i^2$, then the non-cooperative optimal consumption and investment of the $i-th$ asset of the player 1, along with her total wealth are respectively given by

$$c^1(t)=\rho\, W^1(t), \tag{44}$$

$$s_i^1(t)=\frac{\mu_i-\mu_p}{(\sigma_i)^2},\quad i=1,\ldots,p-1, \tag{45}$$

$$W^1(t)=W^1(0)\exp\left[\left(\frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i-\mu_p)^2}{(\sigma_i)^2}+\mu_p-\rho\right)t+\sum_{i=1}^{p-1}\frac{\mu_i-\mu_p}{\sigma_i}\left(B_i(t)-B_i(0)\right)\right]. \tag{46}$$

Proof The proof is very similar to the one of Proposition 1, so we sketch only the different parts. Denoting again by $V_1 \in C^2$ the value function for the player 1 solving (17), we take the utility function as in (43) and assume

$$V_1(W^1(t)) = \frac{\ln(W^1(t))}{A_1} + B_1, \quad A_1, B_1 \in \mathbb{R},\ A_1 \neq 0. \tag{47}$$

Note that here there are two coefficients A_1 and B_1 to be determined. By the first order conditions we get that $s_i^1(t) = (\mu_i - \mu_p)/(\sigma_i)^2$ for $i = 1, \ldots, p-1$, and the consumption $c^1(t) = A_1 W^1(t)$. Moreover, s_i^1 satisfies by hypothesis the condition $0 \leq s_i^1 < \xi_i - s_i^2$. By substituting $c^1(t)$, $s_i^1(t)$ and (47) into Eq. (17) we obtain

$$B_1 = \frac{1}{\rho}\left\{\ln(\rho) + \frac{1}{\rho}\left[\frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i - \mu_p)^2}{(\sigma_i)^2} + \mu_p - \rho\right]\right\},$$

after imposing $A_1 = \rho$ to get rid of $\ln(W^1(t))$. Then, by substituting $A_1 = \rho$ in the expression of $c^1(t)$, we get (44). By replacing (44) and (45) in the first equation of (9) follows the optimal wealth of the player 1:

$$W^1(t) = W^1(0)\exp\left[\left(\frac{1}{2}\sum_{i=1}^{p-1}\frac{(\mu_i - \mu_p)^2}{(\sigma_i)^2} + \mu_p - \rho\right)t + \sum_{i=1}^{p-1}\frac{\mu_i - \mu_p}{\sigma_i}\left(B_i(t) - B_i(0)\right)\right].$$

The transversality condition is

$$\lim_{t\to+\infty} e^{-\rho t}\,\mathbb{E}\left[\frac{\ln(W^1(t))}{A_1} + B_1\right] = \lim_{t\to+\infty}\frac{e^{-\rho t}}{A_1}\mathbb{E}\left[\ln(W^1(t))\right] + \lim_{t\to+\infty} e^{-\rho t} B_1 = 0. \tag{48}$$

Note that the value of the second limit on the right-hand side of (48) is zero, so we focus only on the first limit. Applying Jensen's inequality, we get

$$\mathbb{E}\left[\ln(W^1(t))\right] \leq \ln(\mathbb{E}\left[W^1(t)\right])$$

and $\lim_{t\to+\infty} e^{-\rho t}\ln(\mathbb{E}\left[W^1(t)\right]) = 0$. Hence, by the limit comparison theorem for $t \to +\infty$ follows (48). □

A similar result to Proposition 3 also holds for the player 2.

In the cooperative framework, the following holds.

Proposition 4 *If the utility function takes the form as in (43), then the cooperative optimal consumption and investment of the $i-$th asset of the player $j = 1, 2$, along with her total wealth, are respectively given by*

$$c^j(t) = \rho\, W^j(t), \tag{49}$$

$$s_i^j(t) = \frac{\xi_i}{2}, \quad i = 1, \ldots, p-1, \tag{50}$$

$$W^j(t) = W^j(t) = W^1(0)\exp\left[\left(\frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i - \mu_p) + \mu_p - \rho\right)t + \frac{1}{2}\sum_{i=1}^{p-1}\xi_i\,\sigma_i\,(B_i(t) - B_i(0))\right]. \tag{51}$$

Proof We proceed similarly to Proposition 2 but assuming the utility function as in (43) and the value function solving (31) as

$$V(W^1(t), W^2(t)) = \frac{1}{A}\left(\ln(W^1(t)) + \ln(W^2(t))\right) + B, \quad A, B \in \mathbb{R},\ A \neq 0. \tag{52}$$

The first order condition give us $s_i^j(t) = (\mu_i - \mu_p)/(\sigma_i)^2$ and $c^j(t) = 2\,A W^j(t)$ for $j = 1, 2$. By plugging $s_i^j(t)$, $c^j(t)$ and (52) into (31) we get

$$B = \frac{4}{\rho}\left\{\ln(\rho) + \frac{1}{\rho}\left[\frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i - \mu_p) + \mu_p - \rho\right]\right\},$$

after imposing $A = \rho/2$ to get rid of $\ln(W^j(t))$, and the constraint (13) for every $i = 1, \ldots, p-1$. Using the expression of the cost $c^j(t)$ after replacing $A = \rho/2$, (49), and (50) together with the constraint (13) in the first equation of (9) follows that the optimal wealth of the player j, $j = 1, 2$ is

$$W^j(t) = W^1(0)\exp\left[\left(\frac{1}{4}\sum_{i=1}^{p-1}\xi_i(\mu_i - \mu_p) + \mu_p - \rho\right)t + \frac{1}{2}\sum_{i=1}^{p-1}\xi_i\,\sigma_i\,(B_i(t) - B_i(0))\right].$$

The transversality condition in this case is

$$\begin{aligned}
&\lim_{t\to+\infty} e^{-\rho t}\,\mathbb{E}\left[\frac{1}{A}\Big(\ln(W^1(t)) + \ln(W^2(t))\Big) + B\right] \\
&\quad = \lim_{t\to+\infty}\frac{e^{-\rho t}}{A}\Big(\mathbb{E}\left[\ln(W^1(t))\right] + \mathbb{E}\left[\ln(W^2(t))\right]\Big) \\
&+ \lim_{t\to+\infty} e^{-\rho t} B = 0,
\end{aligned} \tag{53}$$

which holds by applying the same arguments of Proposition 3. □

Remark 3 One can easily observe that the logarithmic utility case could be seen as the limit of the power utility when the constant relative risk aversion tends to 1.

7 Extension to the finite-horizon case

In this section, we consider the finite-horizon formulation of the problem (8), and, as before, we treat both the non-cooperative and cooperative cases. In particular, we report only the values of the optimal controls in both scenarios because their computation is very similar to the one made in Proposition 1 and Proposition 2.

Consider two players, $j = 1, 2$, that invest in the market and share the same number of investments. Both players aim at solving the following finite-horizon problem

$$\begin{aligned}
&\max_{c^j, s_1^j, s_2^j, \ldots, s_{p-1}^j} J(W^j, t, c^j, s_1^j, s_2^j, \ldots, s_{p-1}^j) \\
&:= \mathbb{E}\left[\int_t^T e^{-\rho s}\sum_{j=1}^{2} U_j(c^1(s), c^2(s))ds + S(W^j(T), T)\right],
\end{aligned} \tag{54}$$

subject to the same constraints as in (9).

7.1 The non-cooperative case

Denoting by $\widetilde{V}_1 \in C^{2,1}$ (twice differentiable with respect to $W^1(\cdot)$ and once with respect to t, with continuous derivatives) the value function associated to the problem (54) for the

player 1, the triple $(W^1(t), c^1(t), s_i^1(t))$ is its solution if $\widetilde{V}_1(W^1(t), t)$ solves the following HJB system:

$$\begin{cases} D_t\widetilde{V}_1 + \max\limits_{\{c^1, s_i^1\}} \left\{ \left(\sum\limits_{i=1}^{p-1} \left(s_i^1(t)(\mu_i - \mu_p) + \mu_p\right) W^1(t) - c^1(t) \right) D_{W^1}\widetilde{V}_1 + e^{-\rho t} U_1(c^1(t), c^2(t)) + \right. \\ \qquad \left. \dfrac{1}{2} \sum\limits_{i=1}^{p-1} (s_i^1(t))^2 (\sigma_i)^2 (W^1(t))^2 D^2_{W^1}\widetilde{V}_1 \right\} = 0, \\ \widetilde{V}_1(W^1(T), T) = S(W^1(T), T), \end{cases} \tag{55}$$

where $D_{W^1}\widetilde{V}_1$ and $D^2_{W^1}\widetilde{V}_1$ are defined as in (17), $D_t\widetilde{V}_1$ is the time derivative of $\widetilde{V}_1$, while $S(W^1(T), T)$ is a specified bequest valuation function. Assuming that $U_1(c^1(t), c^2(t))$ is as in (18) and

$$\widetilde{V}_1(W^1(t), t) = \frac{A_1(t)}{1-\alpha_1} e^{-\rho t} (W^1(t))^{1-\alpha_1},\ A_1(t) \in \mathbb{R},\ \alpha_1 > 0,\ \alpha_1 \neq 1, \tag{56}$$

$$S(W^1(T), T) = \frac{\theta_1^{\alpha_1}}{1-\alpha_1} e^{-\rho T} (W^1(T))^{1-\alpha_1},\ \alpha_1 > 0,\ \alpha_1 \neq 1,\ \theta_1 \in (0, 1), \tag{57}$$

we get that $\widetilde{V}_1(W^1(T), T) = S(W^1(T), T)$ if and only if $A_1(T) = \theta_1^{\alpha_1}$.

By the first order conditions, the concavity of $\widetilde{V}_1$ and, recalling that $c^2(t)$ is exogenous for player 1, follows that the optimal consumption and the investment on the i-th asset of the player 1 are

$$c^1(t) = \begin{cases} \left\{\xi_1/[1 + (\xi_1\theta_1 - 1)e^{\xi_1(t-T)}]\right\} W^1(t), & \text{if } \xi_1 > 0, \\ [1/(\theta_1 + T - t)]W^1(t), & \text{if } \xi_1 = 0, \end{cases}$$

$$s_i^1(t) = \frac{\mu_i - \mu_p}{\alpha_1(\sigma_i)^2}, \quad i = 1, \ldots, p-1,$$

where $\xi_1 = A_1$ as defined in (23) and s_i^1 is such that $0 \leq s_i^1 < \xi_i - s_i^2$ for every $i = 1, \ldots, p-1$. For the player 2 in the non-cooperative scenario hold similar results to the ones above, i.e., for $\theta_2 \in (0, 1)$,

$$c^2(t) = \begin{cases} \left\{\xi_2/[1 + (\xi_2\theta_2 - 1)e^{\xi_2(t-T)}]\right\} W^2(t), & \text{if } \xi_2 > 0, \\ [1/(\theta_2 + T - t)]W^2(t), & \text{if } \xi_2 = 0, \end{cases}$$

$$s_i^2(t) = \frac{\mu_i - \mu_p}{\alpha_2(\sigma_i)^2}, \quad i = 1, \ldots, p-1,$$

where ξ_2 is similar to A_1 (unless replacing α_1 with α_2), s_i^2 is such that $0 \leq s_i^2 < \xi_i - s_i^1$ for every $i = 1, \ldots, p-1$.

7.2 The cooperative case

Denoting by $\widetilde{V} \in C^{2,1}$ the value function associated with the problem (54), the triple $(W^j(t), c^j(t), s_i^j(t))$ for $j = 1, 2$ is its solution if $\widetilde{V}(W^1(t), W^2(t), t)$ solves the following HJB system:

$$\begin{cases} D_t\widetilde{V} + \max\limits_{\{c^1,c^2,s_i^1,s_i^2\}} \left\{ \sum\limits_{j=1}^{2} \left\langle \sum\limits_{i=1}^{p-1} \left(s_i^j(t)(\mu_i-\mu_p)+\mu_p\right)W^j(t) - c^j(t), D_{W^j}\widetilde{V} \right\rangle + e^{-\rho t}\, U_1(c^1(t),c^2(t)) + \right. \\ \left. e^{-\rho t}\, U_2(c^1(t),c^2(t))) + \frac{1}{2}\sum\limits_{j,k=1}^{2} \left(\sum\limits_{i=1}^{p-1} (s_i^j(t))(s_i^k(t))(\sigma_i)^2(W^j(t))(W^k(t))D^2_{W^jW^k}\widetilde{V} \right) \right\} = 0, \\ \widetilde{V}(W^1(T),W^2(T),T) = S(W^1(T),W^2(T),T), \end{cases} \tag{58}$$

where $D_{W^j}\widetilde{V}$ and $D^2_{W^j}\widetilde{V}$ are defined as in (31), $D_t\widetilde{V}$ is the time derivative of $\widetilde{V}$, while $S(W^1(T), W^2(T), T)$ is a specified bequest valuation function. Assuming $U_1(c^1(t), c^2(t)) = U_2(c^1(t), c^2(t))$ as in (32) and

$$\widetilde{V}(W^1(t),W^2(t),t) = \frac{A(t)}{1-\alpha}e^{-\rho t}\left[(W^1(t))^{1-\alpha}+(W^2(t))^{1-\alpha}\right],\ A(t)\in\mathbb{R},\ \alpha>0,\ \alpha\neq 1,$$
$$S(W^1(T),W^2(T),T) = \frac{\theta^{\alpha}}{1-\alpha}e^{-\rho T}\left[(W^1(t))^{1-\alpha}+(W^2(t))^{1-\alpha}\right],\ \alpha>0,\ \alpha\neq 1,\ \theta\in(0,1),$$

we get that $\widetilde{V}_1(W^1(T), W^2(T), T) = S(W^1(T), W^2(T), T)$ if and only if $A(T) = \theta^{\alpha}$.

By the first order conditions, the application of the constraint (13) and the concavity of $\widetilde{V}$, follows that the optimal consumption and the investment on the $i-th$ asset of the player j, for $j = 1, 2$, are

$$c^j(t) = \begin{cases} \left\{\beta/\left[1+\left(\frac{\beta\theta}{2^{1/\alpha}}-1\right)e^{\beta(t-T)}\right]\right\}W^j(t), & \text{if } \beta>0, \\ \left[1/\left(\frac{\theta}{2^{1/\alpha}}+T-t\right)\right]W^j(t), & \text{if } \beta=0, \end{cases}$$
$$s_i^j(t) = \frac{\mu_i-\mu_p}{\alpha(\sigma_i)^2},\quad i=1,\ldots,p-1,$$

where $\beta = 2^{1/\alpha}A$, with A as in (38).

8 An illustrative case study: cooperative versus non-cooperative investment strategies

The dynamic and ever-evolving nature of the cryptocurrency market poses unique challenges and opportunities for portfolio management. In this case study, we explore the application of Merton's portfolio management framework, enhanced with a dynamic game approach, to the crypto market, considering the inclusion of a risk-free asset as well. By incorporating game theory principles and considering the strategic interactions among market participants, we aim to gain insights into optimal investment strategies, risk management, and the impact of a risk-free asset in this dynamic and volatile ecosystem. We also use this case study to compare cooperative vs non-cooperative portfolio strategies. We consider a portfolio of five different assets. Data have been collected over the period November 5, 2022–November 5, 2023 from Yahoo Finance. The five different assets considered in our experiment are:

- Fidelity Crypto Industry and Digital Payments ETF (FDIG)
- Bitwise Crypto Industry Innovators ETF (BITQ)
- Bitwise 10 Crypto Index Fund (BITW)
- CMC Crypto 200 Index by Solacti (CMC200)
- Treasury Yield 30 Years (TYX)

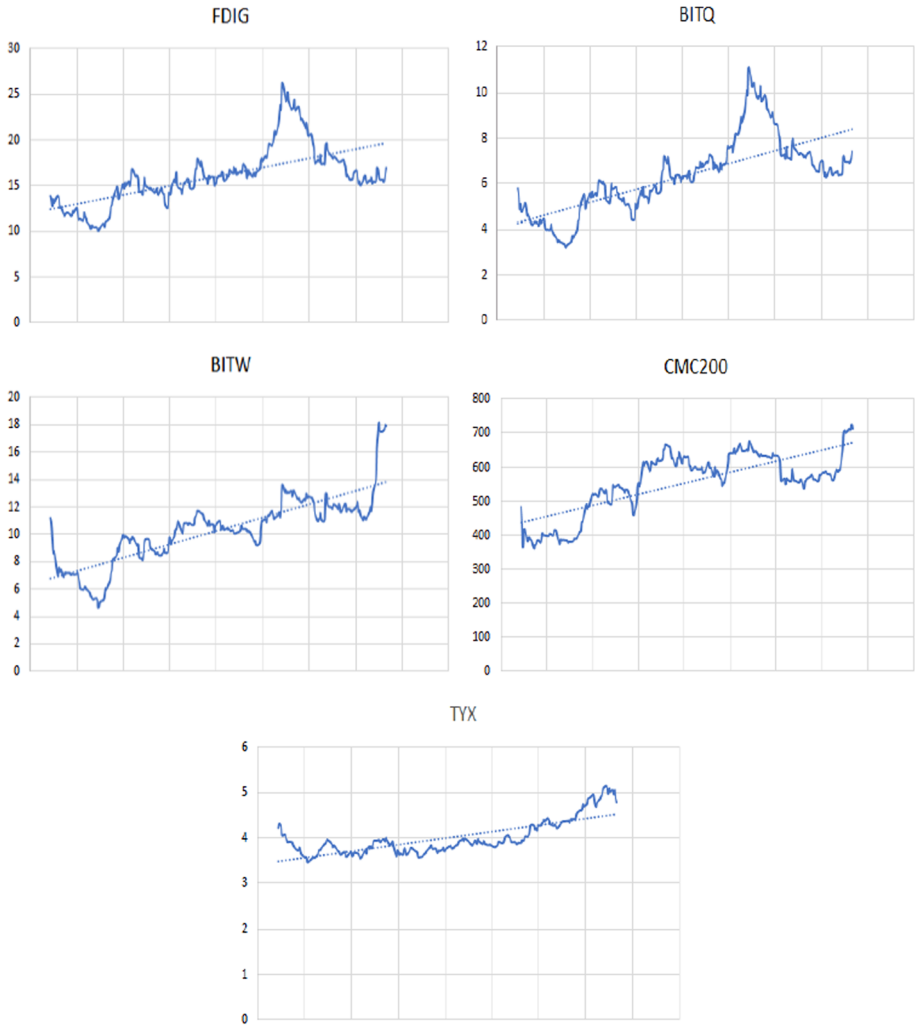

Fig. 1 Daily opening prices temporal evolution

Figure 1 shows the evolution over time of the opening prices of the above assets.
In order to estimate the drift term μ_i and the volatility σ_i, we notice that if we discretise ($\Delta t = 1$ corresponds to one day) the stochastic differential equation of $P_i(t)$ (4), we get that:

$$\frac{P_i(t+1) - P_i(t)}{P_i(t)} = \mu_i + \sigma_i \Delta B_i(t). \tag{59}$$

Now, taking the expectation of both sides of (59), it follows that

$$\mathbb{E}\left(\frac{P_i(t+1) - P_i(t)}{P_i(t)}\right) = \mu_i. \tag{60}$$

At the same time, if we take the variance of both sides, we get

Table 1 Estimated μ_i and σ_i^2

Asset	μ_i	σ_i^2
Fidelity crypto industry and digital payments ETF (FDIG)	$\mu_1 = 0.0015$	$\sigma_1^2 = 0.0014$
Bitwise crypto industry innovators ETF (BITQ)	$\mu_2 = 0.0019$	$\sigma_2^2 = 0.0018$
Bitwise 10 crypto index fund (BITW)	$\mu_3 = 0.0026$	$\sigma_3^2 = 0.0016$
CMC crypto 200 index by solacti (CMC200)	$\mu_4 = 0.0020$	$\sigma_4^2 = 0.0009$
Treasury yield 30 years (TYX)	$\mu_5 = 0.0006$	$\sigma_5^2 = 0.0002$

Table 2 Investment strategies for both players

Asset	Player 1	Player 2	Asset volume
Fidelity crypto industry and digital payments ETF (FDIG)	$s_1 = 0.1428$	$s_1 = 0.0142$	0.1471
Bitwise crypto industry innovators ETF (BITQ)	$s_2 = 0.1604$	$s_2 = 0.0160$	0.1765
Bitwise 10 crypto index fund (BITW)	$s_3 = 0.2777$	$s_3 = 0.0277$	0.3055
CMC crypto 200 index by solacti (CMC200)	$s_4 = 0.3456$	$s_4 = 0.0345$	0.3802
Treasury yield 30 years (TYX)	$s_5 = 0.0735$	$s_5 = 0.9076$	0.9907

$$Var\left(\frac{P_i(t+1) - P_i(t)}{P_i(t)}\right) = \sigma_i^2. \tag{61}$$

Equations (60) and (61) therefore allow to determine the estimation of the drift and the volatility parameters for all different assets.

The results are presented in Table 1. As observed from Fig. 1, these results confirm that investments in TYX are less risky than those in the other assets. For this illustrative example, we consider TYX as a risk-free asset. Moreover, let us assume that only a fixed volume ξ_i of each asset is available on the financial markets, for instance, here, we suppose $\xi_1 = 0.1471$, $\xi_2 = 0.1765$, $\xi_3 = 0.3055$, $\xi_4 = 0.3802$, and $\xi_5 = 0.9907$ millions of dollars. The total volume is, therefore, 2 millions of dollars.

In this financial scenario there are two investors with two different risk aversion parameters, $\alpha_1 = 4.5$ and $\alpha_2 = 45$ as we are considering the case of power utility. Let us notice that the second player has a much higher risk aversion than the first one, so we expect the second player to be more in favour of risk-free investments. We also consider a non-cooperative scenario in which each decision maker $j = 1, 2$, tries to invest in each risky asset an amount s_i^j equal to $(\mu_i - \mu_p)/\alpha_j(\sigma_i)^2$ accordingly to the strategic plan illustrated in Table 2.

This plan clearly shows that both optimal strategies cannot be implemented in the non-cooperative scenario. For instance, the adoption of the optimal strategy from both players would require a volume of the first asset FDIG of 0.1571 million dollars, which is above the current one available on the financial markets. Therefore, in the non-cooperative scenario, one of the two players will have to adopt a sub-optimal strategy. In the cooperative case, instead, the two players share information about the risk and make their own decisions using the same risk aversion parameter. In this case, both players have the chance to put their optimal strategies in place.

9 Managerial implications

The traditional Merton's model has been widely used for dynamic portfolio management, providing valuable insights into optimal investment strategies and risk management. Advancements in the field have led to the development of new formulations that incorporate different approaches, such as multiobjective or dynamic game approaches. A multiobjective approach in portfolio management aims to optimize investment portfolios considering multiple conflicting objectives, such as maximizing return, minimizing risk, and ensuring liquidity. The goal is to identify a set of efficient portfolios that represent different trade-offs between these objectives. A multi-agent framework, instead, focuses on simulating the behaviour and interactions of autonomous agents, such as individual investors or fund managers, within the portfolio management ecosystem. The objective is to understand how the collective actions of these agents impact the overall market dynamics and portfolio performance.

By integrating the dynamic game philosophy into Merton's model, managers gain a more comprehensive understanding of the complex interactions and strategic behaviours within the market. This enables them to make more informed and proactive investment decisions, considering the dynamic nature of the financial landscape. The dynamic game formulation allows for a better assessment and management of risks associated with portfolio management. Managers can analyze the impact of various risk factors and market dynamics on their investment strategies. This understanding enables them to implement more effective risk mitigation measures and hedging strategies, improving portfolio performance and reducing exposure to financial uncertainties.

A dynamic game approach also empowers managers to adapt their investment strategies in response to changing market conditions and the actions of other market participants. The formulation recognizes that investment decisions should not be made in isolation (non-cooperative strategies) but rather as part of a dynamic ecosystem (cooperative strategies). Managers can adjust their portfolios based on the observed behaviours of other investors, anticipating their moves and optimizing their own strategies accordingly.

This new approach can also provide firms with a competitive advantage. By leveraging insights gained from the dynamic interactions and strategic behaviours of market participants, managers can identify investment opportunities that may not be apparent through traditional approaches. This can lead to superior portfolio performance, increased profitability, and a strengthened market position. The dynamic game formulation emphasizes the long-term value-creation potential of dynamic portfolio management. By considering the strategic interactions among investors and incorporating forward-looking, time-dependent information, managers can pursue investment strategies that aim to generate sustainable and consistent returns over time. This approach aligns with the goal of creating long-term value for stakeholders and maximizing shareholder wealth.

10 Conclusion

Starting from the seminal paper by Merton (1969), we have formulated a dynamic game in which two players compete in the same financial market to invest in available p assets. The adoption of a new formulation of Merton's model using a dynamic game-based approach offers significant managerial benefits. It enhances decision-making, improves risk management, enables adaptive strategies, provides a competitive advantage, and fosters long-term value creation. By embracing this innovative approach, managers can navigate the complexi-

ties of the financial markets more effectively, leading to enhanced portfolio performance and, ultimately, organizational success. To simplify the presentation we have supposed to consider $p-1$ risky assets and only one risk-free asset. We have discussed both the infinite and the finite-horizon cases under cooperative and non-cooperative strategies. By assuming a CRRA utility function, we determine the closed-form solution and the optimal time-dependent portfolio allocation among different assets. By comparing the optimal strategies of the cooperative case with those of the non-cooperative case, one can conclude that the optimal scenario can be attained only if a cooperative strategy is put in place. Of course, these results are based on specific model assumptions that allow us to determine a closed-form solution. However, a real-world situation involves more complicated factors, such as discrete amounts of assets to hold, discrete quantities of trades, transaction costs, non-stationary and correlated processes of multiple assets, uncertain risk-free rate, consumption constraints, arbitrary risk aversion, and so on. In conclusion, the model sheds light on the optimal interaction between different financial players. Future directions include the extension of the dynamic game approach to different utility functions as well as the use of different stochastic processes for risky assets.

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Declaration

Conflict of interest The authors declare no Conflict of interest.

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